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# HILLSBOROUGH COUNTY PARCEL SALES PREDICTION

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## Abstract

This study explores the use of machine learning to predict residential property sale values in Hillsborough County, Florida. Accurate valuation is critical for disaster response and recovery planning, particularly in post-hurricane damage assessments. The project involved extracting and merging parcel and historical sales data, followed by cleaning, feature selection, and exploratory data analysis for residential parcel sales from 2014-2024. A Box-Cox transformation was applied to correct for target variable skewness, and models were trained using a consistent pipeline that handled preprocessing for the baseline models. Five models were evaluated: Linear Regression, Elastic Net, Random Forest, Artificial Neural Network (ANN), and Light Gradient Boosting Machine (LightGBM). The models were optimized by randomly sampling hyperparameters from a parameter grid and fine tuning the grid after each iteration. Model performance was assessed using root mean squared error (RMSE) in the transformed scale for the training subset and USD on a holdout testing subset. Ensemble models, particularly Random Forest and LightGBM, outperformed linear methods, capturing complex, non-linear relationships in the data. The final models demonstrate the feasibility of using machine learning for scalable, data-driven property valuation in support of emergency management applications.

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# 1 Introduction

When a natural disaster, such as a hurricane, impacts a community, government entities rely on metrics from property appraisers to quantify and categorize the damages. For example: *Market Value* is the value that the land and improvements would sell for in a competitive market at a fair price; *Building Value* is the value of all structures on a property; and *Tax Value* is the assessed value minus any taxable exemptions. An issue with these metrics is that they are not standardized in how they are calculated or how they are used by local governments, and the *Market Value* may not accurately represent price a buyer would pay to purchase the property. In addition, programs such as the Save Our Homes Act introduces bias by limiting the annual increase in the assessed value of homesteaded properties to 3% or the change in the Consumer Price Index (CPI), whichever is less (Florida Legislature, 2025). As a result, estimating total impacts can be challenging, and reimbursement programs may undervalue properties leaving substantial financial burdens on the residents while filing an insurance claim.

Prior research highlights a clear shift from traditional hedonic regression toward more flexible machine learning methods. The University of Colorado at Boulder found Artificial Neural Network and Random Forest models outperformed traditional models in valuing property sales in Boulder, Colorado (Yazdani, 2021). Similarly, others have demonstrated that boosting models like Light Gradient Boosting Machine were more robust than bagging and linear methods, especially across structural market changes during COVID-19 due to the ability to capture non-linear trends (Mora-Garcia, Cespedes-Lopez, & Perez-Sanchez, 2022). Model selection and tuning when applying regression-based methods to housing data has risks of overfitting when analyzing smaller sample sizes and it's beneficial to compare the performance of ensemble models with simple models such as Linear Regression when dealing with property sales (Viktorovich, Aleksandrovich, Leopoldovich, & Vasilevna, 2018).

This study builds directly on these insights by comparing an expanded set of models, tuning hyperparameters, and adjusting for skewed sales via Box-Cox to deliver a more resilient predictive framework for Hillsborough County, Florida.

## 2 Methodology

Data from the Hillsborough County Property Appraiser's website containing metrics about the parcels, buildings, and sale values is loaded into an environment and analyzed during the exploratory data analysis (EDA) phase to uncover hidden trends, multicollinearity, outliers, and other statistically significant factors. A cleaned version of the data is split into training and testing subsets and will use a Box-Cox transformation to transform the data using the optimal  $\lambda$  prior to model training. The training subset undergoes multiple rounds of machine learning and deep learning hypertuning using a randomized search with cross-validation to identify the optimal hyperparameters based on the lowest cross-validation error. Once the optimal

hyperparameters are determined, the models is fitted to the testing data subset to measure its performance using the root mean squared error shown in the formula below:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

See Appendix A for an analytical workflow diagram.

## 2.1 Models

This analysis considers 5 models: Linear Regression, Elastic Net, Random Forest, Artificial Neural Network, and Light Gradient Boosting Machine. Linear Regression and Elastic Net provide strong baseline approaches and offer interpretability, while Random Forest and Light Gradient Boosting Machine are well suited for capturing non-linear relationships. The Artificial Neural Network was included to evaluate complex patterns through a deep learning approach. These models were selected to represent a diverse range of predictive approaches and balances interpretability and complexity using machine learning and deep learning.

### 2.1.1 Linear Regression

A foundational statistical model that fits a straight line to the data by minimizing the sum of squared differences between observed and predicted values. It is ideal for predicting continuous outcomes that exhibit a linear relationship with input features and is valued for its simplicity and interpretability.

Linear Regression can be described as the following equation:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$$

- $\hat{y}$ : Predicted value
- $\beta_0$ : Intercept
- $\beta_j$ : Coefficient for the  $j$ 'th feature
- $x_j$ : Input of the  $j$ 'th predictor
- $n$ : Number of features

### 2.1.2 Elastic Net

A regularized Linear Regression model that blends the strengths of Lasso (L1) and Ridge (L2) penalties. It connects input features directly to the output through weighted linear combinations, optimizing these weights during training to minimize both prediction error and a combined regularization term. Elastic Net is especially effective for high-dimensional datasets with multicollinearity, offering a balance between feature selection and model stability.

Elastic Net can be described by minimizing the following equation:

$$\hat{\beta} = \min_{\beta} ||(y_i - \hat{y}_i)||^2 + \lambda_1 \sum_{j=1}^p |\beta_j| + \lambda_2 \sum_{j=1}^p \beta_j^2$$

- $y_i$ : Observed value for instance  $i$
- $\hat{y}_i$ : Predicted value for instance  $i$
- $n$ : Number of observations
- $\beta_j$ : Coefficient for feature  $j$
- $\lambda_1$ : L1 regularization strength (Lasso)
- $\lambda_2$ : L2 regularization strength (Ridge)
- $p$ : Number of features

### 2.1.3 Random Forest

A bagging ensemble learning method that constructs multiple decision trees using bootstrapped samples and random subsets of features. Each tree contributes to the final prediction through majority voting (classification) or averaging (regression). Random Forests are robust to overfitting and perform well on complex, high-dimensional datasets with nonlinear relationships.

There is no single close-formed solution for Random Forest, but it can typically be described as drawing a bootstrap sample from the training dataset and fitting a model to each sample as shown in the equation below:

$$\hat{y}(x) = \frac{1}{B} \sum_{b=1}^B h_b(x)$$

- $\hat{y}(x)$ : Final prediction at point  $x$
- $h_b(x)$ : Prediction from the  $b$ 'th decision tree
- $B$ : Total number of trees

### 2.1.4 Light Gradient Boosting Machine (LightGBM)

A high-performance gradient boosting framework that builds decision trees sequentially, with each new tree correcting the errors of its predecessors. It uses histogram-based algorithms and leaf-wise tree growth to improve speed and accuracy. LightGBM is well-suited for large-scale structured data and excels in both classification and regression tasks, especially when model performance is prioritized over training time.

Like Random Forest, LightGBM is not a single close-formed equation, but the model is built iteratively as described in the equation below:

$$\hat{y}^t = \hat{y}^{t-1} + \eta \cdot h_t(x)$$

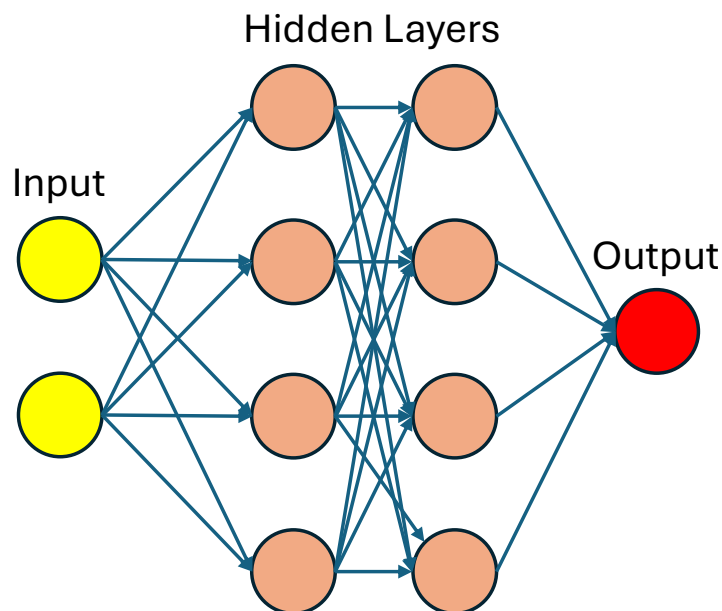
- $\hat{y}^{(t)}$ : Prediction from iteration  $t$
- $\hat{y}^{(t-1)}$ : Prediction from previous iteration
- $h_t(x)$ : New decision tree at iteration  $t$
- $\eta$ : Learning rate

### 2.1.5 Artificial Neural Network (ANN)

A layered computational model composed of input, hidden, and output layers, where each node simulates a neuron. During training, the network uses an activation function and adjusts the weights of connections between nodes to minimize prediction error. ANNs are powerful for capturing intricate patterns and nonlinear dependencies, making them suitable for tasks like image recognition, time series forecasting, and natural language processing.

For a simple neural network with hidden layers, ANN can be described as the following diagram with 2 input layers and 1 output layer:

Figure 1: ANN Diagram



## 3 Exploratory Data Analysis

Raw geographical information systems (GIS) data for all property sales and parcel details as of April 2025 are read into `pandas` dataframes and merged based on a primary key (folio) while dropping non-analytical fields. In the event of duplicate fields, values from the sales dataframe are preserved. The schema is listed below.

Table 1: Schema for Finalized Data

| Feature  | Data Type | Description                       |
|----------|-----------|-----------------------------------|
| FOLIO    | object    | Folio of parcel. Primary key      |
| SITE_ZIP | category  | Postal zip code                   |
| tBEDS    | float64   | Number of beds                    |
| tBATHS   | float64   | Number of baths                   |
| tSTORIES | float64   | Number of stories                 |
| tUNITS   | float64   | Number of units                   |
| tBLDGS   | float64   | Number of buildings               |
| HEAT_AR  | float64   | Heating area in ft <sup>2</sup>   |
| ACREAGE  | float64   | Acreage of lot                    |
| LU_GRP   | category  | Generalized land use group        |
| S_YEAR   | category  | Sale year derived from sale date  |
| S_AMT    | float64   | Sale Amount (USD). Response value |

### 3.1 Land Use Group

The majority of property sales records occur for residential properties (87.26%, n= 2,053,262). This analysis focuses on single-family and multi-family properties to reduce potential bias from the imbalance in property types. The table below shows the distribution by land use group.

Table 2: Land Use Breakdown for Raw Data

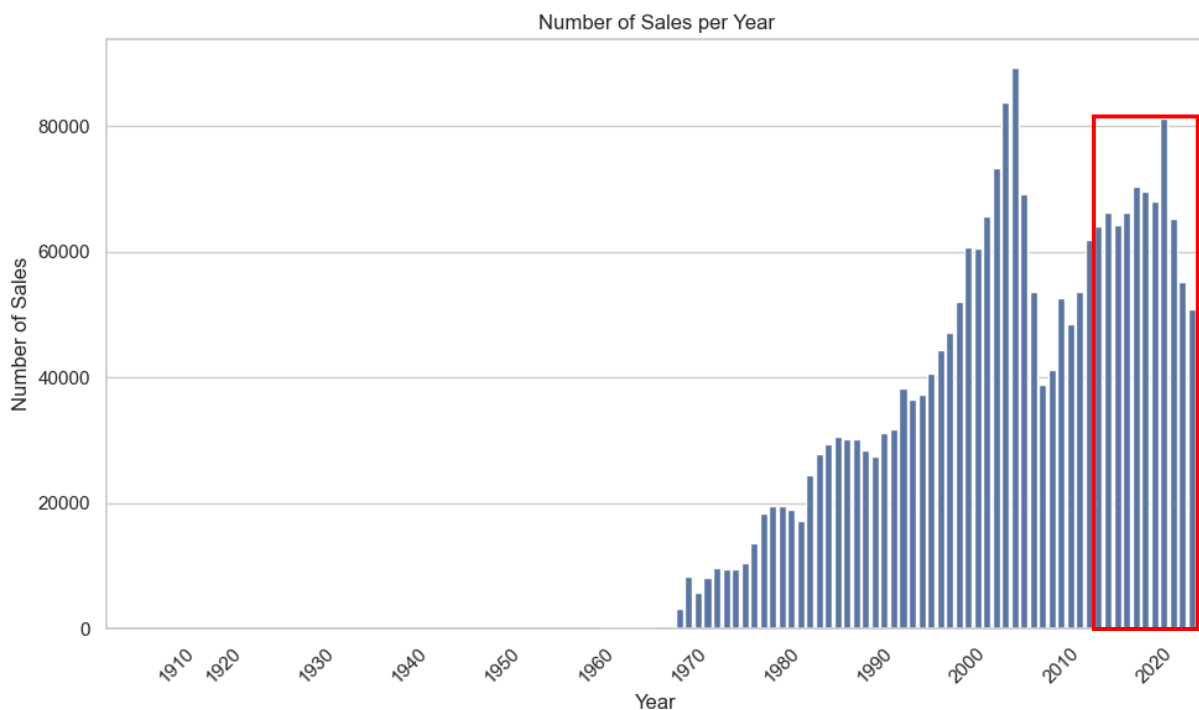
| Land Use Group                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Count            | %            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|--------------|
| <b>SINGLE FAMILY</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>1,811,627</b> | <b>76.99</b> |
| <b>MULTI-FAMILY</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>241,635</b>   | <b>10.27</b> |
| VACANT RESIDENT                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 70,837           | 3.01         |
| MOBILE HOMES                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 62,889           | 2.67         |
| COMM./OFFICE                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 35,423           | 1.51         |
| OTHER: (INVALID CODE, AGRICULTURAL, GOVERNMENT, STORE/SHP CENTE, INSTITUTIONAL, LIGHT INDUS., VACANT COMM., VACANT TOWNHOME, UTILITY, R/W, OFFICE CONDO, SCHOOLS/COLLEGE, VACANT INDUS., HEAVY INDUS., MEDICAL OFFICE CONDO, APT CONDO, COMMERCIAL, INDUSTRIAL CONDO, CHURCH PARSONAGES, COOPERATIVE, 1 STY STORE CONDO, VACANT MH/COOP, SFR BLD ARND MH, MINING, MARINAS, CEMETERY, PRIVATE GC, SEMI PRIVATE GC, LANDSCAPE/WALL BUFFER, PARKS, HOA COMMERCIAL, EXEC/PRACTICE GC | 130,731          | 5.56         |



## 3.2 Time Frame

The Hillsborough County Property Appraiser provides digital records of property sales dating back to 1901; however, the quantity of records doesn't start showing a significant amount until the late 1960s. The state of the economy also affects real estate valuation and can play a significant factor in the ability to purchase a home due to the cost of construction materials and the ability to acquire a loan (Duca, 2013). To avoid significant financial events such as the Subprime Mortgage Crisis which eventually led to a housing market crash in 2008, records from 2014-2024 are used for this analysis. The figure below shows the distribution of sales by year.

Figure 2: Sales Distribution for Raw Data



## 3.3 Outliers

Outliers and sentinel values, such as low-dollar family transactions, are systematically removed using a minimum sales value of \$50,000 and an Isolation Forest. This model is an anomaly detection algorithm that iteratively isolates observations by randomly selecting features and a split value until the instance is isolated or the tree reaches a predefined height limit and scores the average path length. The key concept is that anomalies are easier to isolate because they require fewer random splits to be separated from the rest of the data.

The scoring for an Isolation Forest can be described as below:

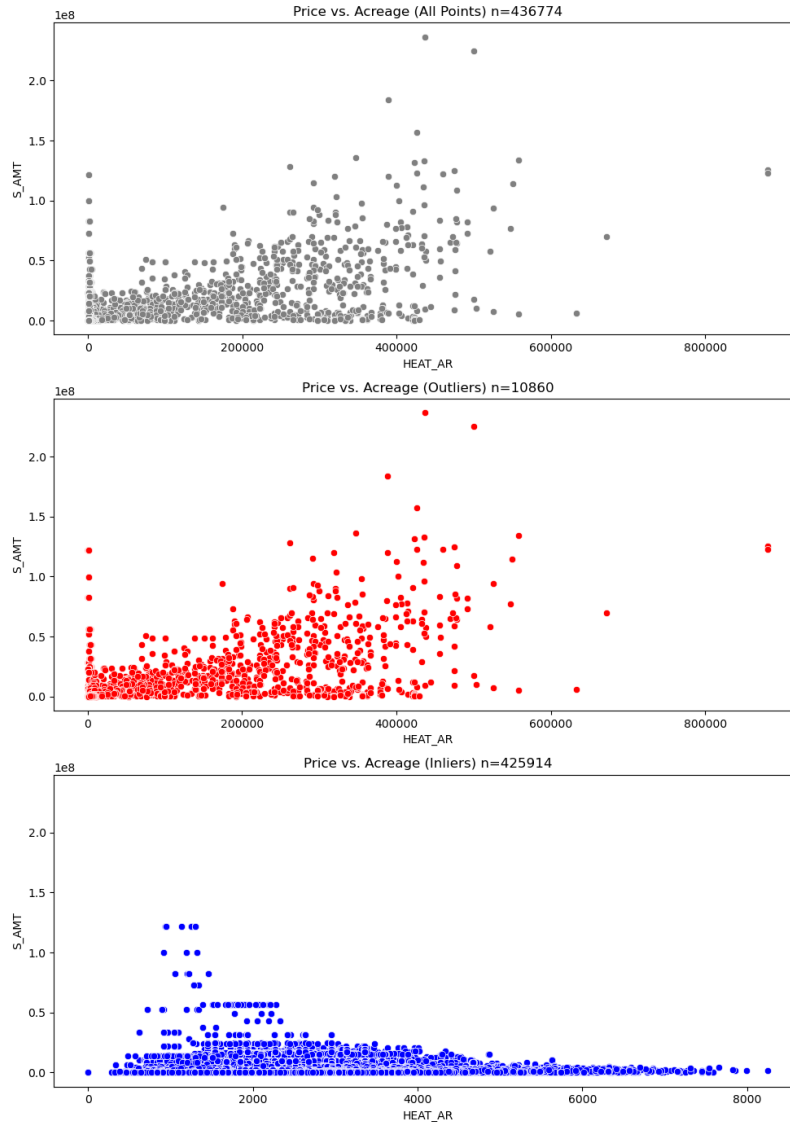
$$s(x, n) = 2 \frac{E(h(x))}{c(n)}$$

$$c(n) = 2H(n-1) - \frac{2(n-1)}{n}$$

- $s(x, n)$ : anomaly score for instance  $x$  in dataset of size  $n$
- $E(h(x))$ : average path length of instance  $x$  across all trees
- $c(n)$ : average path length of unsuccessful search in binary tree on  $n$  samples
- $H(i)$ : harmonic number; approximated as  $\ln(i) + \gamma$

The scores can be interpreted as  $s(x) \approx 1$  is likely an anomaly, and  $s(x) < 0.5$  is likely normal point. A `scikit-learn` Isolation Forest with contamination value of 0.025, number of estimators as 250, and max sample size of 256 was fitted to the data by year and the results are shown in the figure below.

Figure 3: Outlier Removal

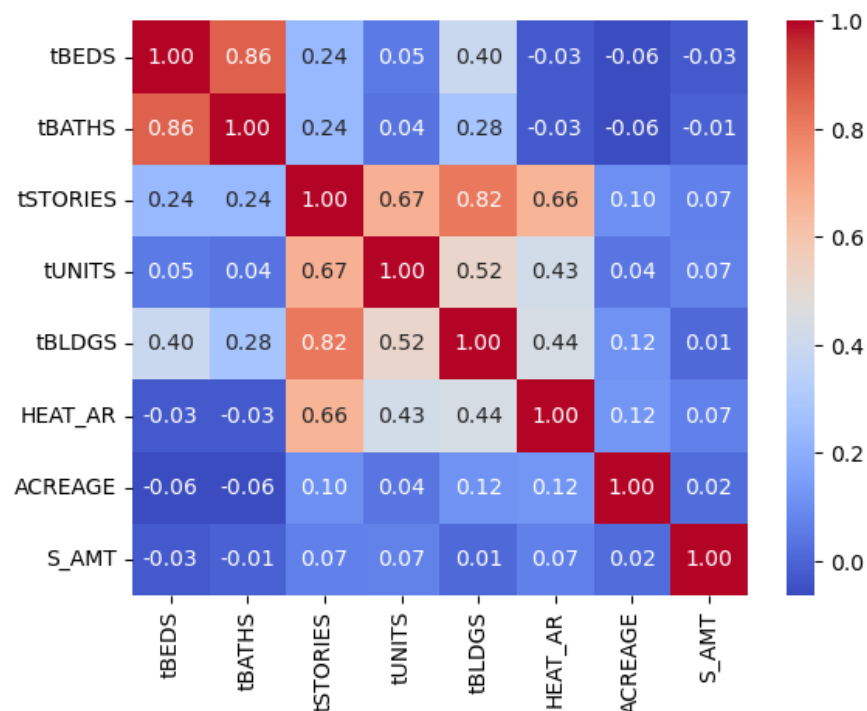


Heating area (ft<sup>2</sup>) was used for comparison to graphically represent the outlier/inlier relationships and trends from the Isolation Forest. This captured both sentinel values in the lower tail and extreme high-value records in the upper tail which are absent in the [Inliers](#) plot. Approximately 11,000 data points were removed from 2014–2024 as described on the [Outliers](#) and [Inliers](#) plots.

### 3.4 Multicollinearity

Before training predictive models, it is important to assess multicollinearity among the input features to ensure stability and reliability. Multicollinearity occurs when two or more independent variables are highly correlated, which can distort model coefficients, inflate standard errors, and reduce interpretability. While Elastic Net and some other models are more reliant to multicollinearity, this can especially be problematic for simple linear models. To evaluate this, a correlation heatmap was used to visualize pairwise relationships among numeric features. Additionally, a Variance Inflation Factor (VIF) analysis was performed to measure how much the variance of an estimated regression coefficient is inflated due to multicollinearity.

Figure 4: Correlation Plot



There is a strong correlation between number of beds to number of baths (0.86) and number stories to number of buildings (0.82), and all predictors have a weak correlation to the response sales price.

Table 3: VIF Results

| Feature  | VIF  |
|----------|------|
| tSTORIES | 6.24 |
| tBEDS    | 5.03 |
| tBATHS   | 4.59 |
| const    | 4.20 |
| tBLDGS   | 4.04 |
| HEAT_AR  | 1.96 |
| tUNITS   | 1.89 |
| ACREAGE  | 1.04 |

No predictors have a high Variance Influence Factor ( $VIF \geq 10$ ) while the number of stories and number beds have a moderate VIF.

### 3.5 Transformation

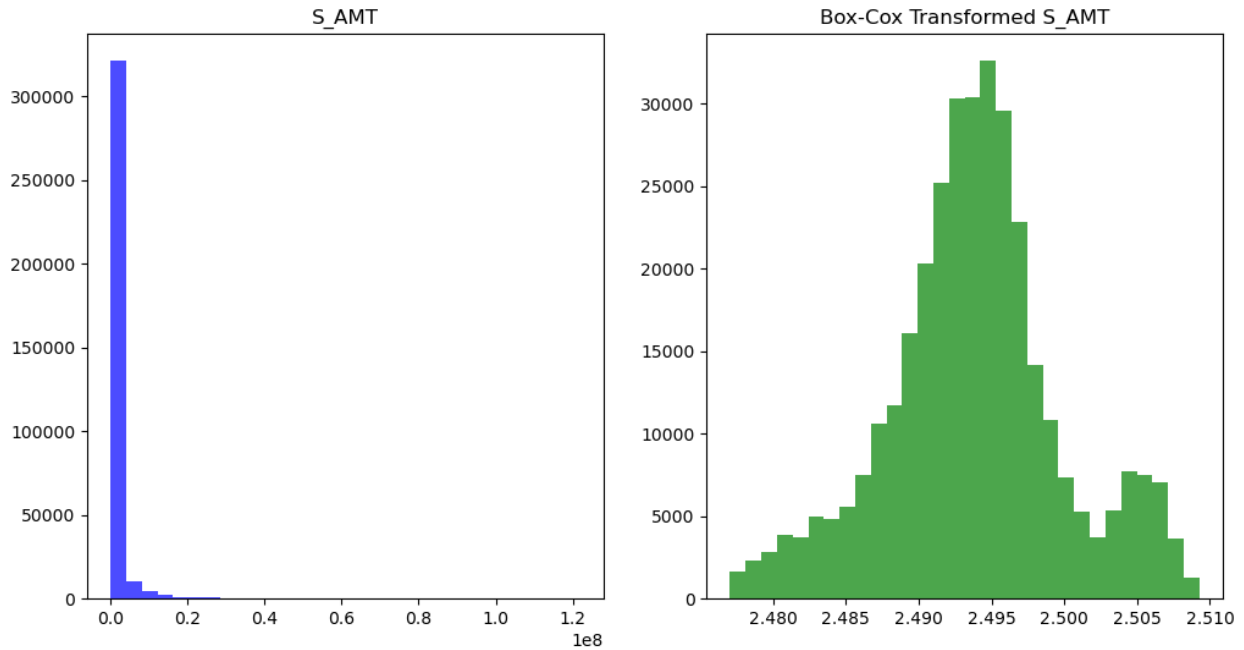
One-hot encoding is a method to convert categorical data into numerical format by representing the values as binary vectors. This ensures models treat each category as independent and avoids an implicit ordering system. Zip codes and land use group values were one-hot encoded to allow the algorithms to understand it and use it efficiently.

When analyzing datasets using machine learning, a normal distribution is often assumed. If ignored, the results can lead to biased parameter estimation, inaccurate confidence intervals, and faulty hypothesis testing. For this dataset, sales amounts do not follow a normal distribution and needs to be transformed prior to training. The training data's sales amount is run through a Box-Cox transformation and the response is transformed using the model's optimal  $\lambda = -0.39828$  parameter. The Box-Cox transformation formula equation and the distributions before and after transformation are shown below.

$$y^{(\lambda)} = \frac{y^\lambda - 1}{\lambda}$$

- $y$ : original data
- $y^{(\lambda)}$ : transformed data
- $\lambda$ : transformation control value

Figure 5: Box-Cox Transformation



## 4 Analysis

### 4.1 Training

The final dataset of 425,914 records is randomly split into training (80%,  $n = 340,731$ ) and testing (20%,  $n = 85,183$ ) subsets. To establish a fair and consistent foundation for model comparison, a baseline pipeline was established for each algorithm and included preprocessing steps such as standardization of numerical features and one-hot encoding for categorical features. This ensured all models received the same transformation and avoided data leakage during cross-validation.

Each model's pipeline was hypertuned using `RandomizedSearchCV`, which randomly samples from a predefined parameter grid. This approach provides a more computationally efficient approach compared to an exhaustive grid search while still exploring a wide range of hyperparameter options. The tuning process was performed using a 5-fold cross-validation and evaluated based on the RMSE to identify the best combination of hyperparameters for each model. Once the best hyperparameters were selected, the parameter grid was adjusted to fine tune each model until the optimal hyperparameters were discovered. Consideration was given to inputs such as number of estimators or tree depth to avoid potentially overfitting the training data.

Table 4: Parameter Grid

| Model                     | Hyperparameters                                                                                                                               | Optimal Hyperparameters                                                                                               |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Linear                    | None                                                                                                                                          | Use default                                                                                                           |
| Elastic Net               | $\alpha$ , L1 ratio                                                                                                                           | $\alpha=0.001$ , L1=0.01                                                                                              |
| Random Forest             | Number of estimators, max tree depth                                                                                                          | max depth=25, number of estimators=425                                                                                |
| LightGBM                  | Number of estimators, initial learning rate, number leaves, min child samples, subsample fraction, boosting sample fraction, L1 term, L2 term | learning rate=0.05, min child samples=10, n estimators=300, num leaves=56, L2=0.5, subsample=0.9                      |
| Artificial Neural Network | Hidden layer size, $\alpha$ , initial learning rate, max iterations, activation function                                                      | $\alpha=0.001$ , hidden layer sizes=(64,), initial learning rate=0.0005, max iterations=300, activation function=relu |

## 4.2 Results

For each model, the hyperparameters that achieved the lowest cross-validation error were used to fit and evaluate the best models on the training and testing data. To evaluate the models, RMSE was calculated during training on the Box-Cox–transformed target, and predictions on the testing subset were inverse-transformed to compute RMSE on the original sales amount scale of USD. The transformed RMSE reflects how well the model learned from a normalized distribution while the original scale RMSE reflects the error in real monetary terms for practical interpretation. Reporting RMSE on both the transformed and original scale provides a more complete picture of model performance and provides both statistical validity and stakeholder relevance. Random Forest performed the best with the lowest RMSE while ANN had the highest error of all models evaluated.

Table 5: RMSE Results

| Model         | RMSE (training, $\lambda$ transformed) | RMSE (testing, USD) |
|---------------|----------------------------------------|---------------------|
| Linear        | 0.0050849                              | 2,419.10            |
| Elastic Net   | 0.0051152                              | 2,419.77            |
| Random Forest | 0.0044144                              | 2,382.29            |
| LightGBM      | 0.0045119                              | 2,406.30            |
| ANN           | 0.0051727                              | 2,419.82            |

## 5 Conclusion

This project aimed to develop an accurate model to predict single-family and multi-family property sale values in Hillsborough County, Florida from 2014-2024, with the goal of supporting post-disaster damage assessments. This analysis involved a complete pipeline from extracting and cleaning property appraiser and historical sales data to feature engineering, exploring relationships, and applying advanced analytical techniques through machine learning and deep learning.

The quality of an analysis using property appraiser data heavily relies on outlier detection. This data is largely manually entered which can result in data entry errors as well as missing data points represented by sentinel values as placeholders and extreme low values during changes in property ownership within the same family. In this study, outlier detection was accomplished using a conservative and realistic minimum threshold of \$50,000 and an Isolation Forest, which helped identify and remove anomalous placeholder records by recursively isolating and splitting the data. By cleaning the data of such inconsistencies, the models were better able to generalize and produce meaningful results.

A Box-Cox transformation was necessary to normalize the skewed response variable, sales amount, to improve model performance and stability. Analytical models were trained and evaluated, including Linear Regression, Elastic Net, Random Forest, ANN, and LightGBM. Each model was processed through a consistent pipeline and parameter grid that handles preprocessing and hyperparameter optimization using randomization with cross-validation.

Among the models used, ensemble methods like Random Forest and Light Gradient Boosting Machine outperformed others in terms of RMSE when evaluated on the holdout testing subset. While simple baseline models such as Linear Regression provide transparency and interpretability, boosting and bagging models better captured the complex and non-linear relationships shown in the data.

The final models showed predictive accuracy at levels suitable for offering valuable insights for future property valuation for emergency response planning and recovery efforts. The analysis could be further improved by incorporating a quality indicator or condition rating into the set of predictor variables. While structural characteristics such as the number of bedrooms and bathrooms provide a baseline for valuation, they do not fully capture differences in structural condition or renovation status, and properties of similar size and amenities can vary significantly in value.

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## Appendix A: Analytical Flowchart



# Hillsborough County Parcel Sales Analysis

